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Filtering apparatuses and systems for electronic enclosures

Abstract

An apparatus for attaching to an electronic unit includes a base and a cover. The base includes a bottom wall defining an opening, a first sidewall extending from a perimeter of the bottom wall, a support wall, and a second sidewall. The first sidewall and the second sidewall extend from respective ends of the support wall such that a groove is defined between the first sidewall, the support wall, and the second sidewall. The cover is configured to be coupled to the base and includes a top wall including a first side and a second side, a third sidewall extending from a perimeter of the second side of the top wall, and at least one spacer extending from the second side of the top wall and configured to define a cavity between the base and the cover.

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Background/Summary

RELATED APPLICATION

(1) This application claims benefit of priority from Finnish Patent App. No. 20235864, filed Aug. 3, 2023, the disclosure of which is hereby incorporated in its entirety by reference herein.

TECHNICAL FIELD

(2) One or more example embodiments relate to filtering apparatuses and systems for electronic enclosures.

BACKGROUND

(3) Enclosures for electrical units include openings or vents that allow air to enter and exit the enclosure. Such openings or vents may become blocked, such as by liquid and/or dust, preventing air flow and creating pressure within the enclosure, which may cause the electronic unit to malfunction.

SUMMARY

(4) At least one example embodiment relates to an apparatus for attaching to an electronic unit. The apparatus includes a base and a cover. The base includes a bottom wall defining an opening, a first sidewall extending from a perimeter of the bottom wall, a support wall, and a second sidewall. The first sidewall and the second sidewall extend from respective ends of the support wall such that a groove is defined between the first sidewall, the support wall, and the second sidewall. The cover is configured to be coupled to the base and includes a top wall including a first side and a second side, a third sidewall extending from a perimeter of the second side of the top wall, and at least one spacer extending from the second side of the top wall and configured to define a cavity between the base and the cover.

(5) In at least one example embodiment, the at least one spacer includes a protrusion extending from a center of the second side of the top wall and a plurality of ribs extending from the second side of the top wall between the protrusion and the third sidewall. The plurality of ribs are configured to contact at least a top surface of the support wall of the base.

(6) In at least one example embodiment, an interior surface of the third sidewall of the cover, the second side of the top wall of the cover, an exterior surface of the second sidewall, and a top surface of the support wall define a first fluid pathway. An interior surface of the third sidewall of the cover, the second side of the top wall of the cover, an exterior surface of the second sidewall, a top surface of the support wall, and an interior surface of the first sidewall define a second fluid pathway.

(7) In at least one example embodiment, the first fluid pathway is a liquid pathway configured to direct liquid away from the electronic unit and the second fluid pathway is an air pathway in fluid communication with the electronic unit.

(8) In at least one example embodiment, the apparatus includes an air vent within the opening of the bottom wall.

(9) In at least one example embodiment, the air vent is configured to prevent passage of liquid and solid particles into the electronic unit.

(10) In at least one example embodiment, the cover is configured to cover the air vent and the at least one spacer is configured to contact at least a portion of a top surface of the air vent.

- (11) In at least one example embodiment, the top wall of the cover defines at least one hole. The at least one hole is configured to receive at least one attachment device for securing the cover to the base.
- (12) In at least one example embodiment, the at least one attachment device includes one or more screws.
- (13) In at least one example embodiment, the first sidewall and the second sidewall are perpendicular to the support wall.
- (14) In at least one example embodiment, a length of the first sidewall is greater than a length of the second sidewall.
- (15) At least one example embodiment relates to a filtering system for an electronic unit. The filtering system includes a base, an air vent, and a cover. The base includes a bottom wall defining an opening, a first sidewall extending from a perimeter of the bottom wall, a support wall, and a second sidewall. The first sidewall and the second sidewall extend from respective ends of the support wall such that a groove is defined between the first sidewall, the support wall, and the second sidewall. The air vent includes a first portion configured to extend through the opening in the bottom wall of the base and a second portion at an end of the first portion. The first portion defines a channel, and the second portion is in fluid communication with the channel. The cover is configured to be coupled to the base and cover the air vent. The cover includes a top wall including a first side and a second side, a third sidewall extending from a perimeter of the second side of the top wall, and at least one spacer extending from the second side of the top wall and configured to define a cavity between the base and the second side of the cover.
- (16) In at least one example embodiment, the at least one spacer includes a protrusion extending from a center of the second side of the top wall and configured to contact a top surface of the air vent and a plurality of ribs extending from the second side of the top wall between the protrusion and the third sidewall. The plurality of ribs are configured to contact at least a top surface of the support wall of the base.
- (17) In at least one example embodiment, the second portion of the air vent includes a top part and a bottom part. The bottom part is at the end of the first portion of the air vent and the top part is separated from the bottom part by a plurality of supports extending between the top part and the bottom part.
- (18) In at least one example embodiment, a space defined between the top part and the bottom part is in fluid communication with the channel.
- (19) In at least one example embodiment, an interior surface of the third sidewall of the cover, the second side of the top wall of the cover, an exterior surface of the second sidewall, and a top surface of the support wall define a first fluid pathway. An interior surface of the third sidewall of the cover, the second side of the top wall of the cover, an exterior surface of the second sidewall, a top surface of the support wall, an interior surface of the first sidewall, the second portion of the air vent, and the channel of the first portion of the air vent define a second fluid pathway.
- (20) In at least one example embodiment, the first fluid pathway is a liquid pathway configured to direct liquid away from the electronic unit and the second fluid pathway is an air pathway in fluid communication with the electronic unit.
- (21) In at least one example embodiment, the filtering system includes a gasket surrounding at least a portion of the first portion of the air vent. An exterior surface of the first portion of the air vent includes a plurality of threads.
- (22) In at least one example embodiment, the air vent is configured to prevent passage of liquid and solid particles into the electronic unit.
- (23) In at least one example embodiment, the top wall of the cover defines at least one hole. The at least one hole is configured to receive at least one attachment device for securing the cover to the base.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The various features and advantages of the non-limiting embodiments herein may become more apparent upon review of the detailed description in conjunction with the accompanying drawings. The accompanying drawings are merely provided for illustrative purposes and should not be interpreted to limit the scope of the claims. The accompanying drawings are not to be considered as drawn to scale unless explicitly noted. For purposes of clarity, various dimensions of the drawings may have been exaggerated.
- (2) FIG. 1A is a perspective view of an electronic unit according to at least one example embodiment.
- (3) FIG. 1B is a detailed, perspective view of a filtering system of the electronic unit of FIG. 1A according to at least one example embodiment.
- (4) FIG. 2 is a perspective view of the filtering system of FIG. 1B according to at least one example embodiment.
- (5) FIG. 3 is an exploded view of the filtering system of FIG. 2 according to at least one example embodiment.
- (6) FIG. 4A is a top, perspective view of a base of the filtering system of FIGS. 2-3 according to at least one example embodiment.
- (7) FIG. 4B is a bottom, perspective of the base of FIG. 4A according to at least one example embodiment.
- (8) FIG. 4C is a side, cross-section view of the base of FIG. 4A according to at least one example embodiment.
- (9) FIG. 5A is a top, perspective view of a cover of the filtering system of FIGS. 2-3 according to at least one example embodiment.
- (10) FIG. 5B is a bottom, perspective view of the cover of FIG. 5A according to at least one example embodiment.
- (11) FIG. 6A is a side, perspective view of an air vent of the filtering system of FIGS. 2-3 according to at least one example embodiment.
- (12) FIG. 6B is a side, cross-section view of the air vent of FIG. 6A according to at least one example embodiment.
- (13) FIG. 7A is a side, cross-section view of a base and a cover of FIG. 2 according to at least one example embodiment.
- (14) FIG. 7B is a side, cross-section view of the filtering system of FIG. 2 according to at least one example embodiment.
- (15) FIG. 8A is a detailed, cross-section view of the filtering system and the electronic unit of FIGS. 1A-1B showing a first fluid pathway according to at least one example embodiment.
- (16) FIG. 8B is a detailed, cross-section view of the filtering system and the electronic unit of FIGS. 1A-1B showing a second fluid pathway according to at least one example embodiment.
- (17) FIG. 9 is a detailed, cross-section view of another filtering system and electronic unit according to at least one example embodiment.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

- (18) Some detailed example embodiments are disclosed herein. However, specific structural and functional details disclosed herein are merely representative for purposes of describing some example embodiments. Example embodiments may, however, be embodied in many alternate forms and should not be construed as limited to only example embodiments set forth herein.
- (19) Accordingly, while example embodiments are capable of various modifications and alternative forms, example embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that there is no intent to limit an example

embodiment to the particular forms disclosed, but to the contrary, example embodiments are to cover all modifications, combinations, equivalents, and alternatives falling within the scope of an example embodiment. Like numbers refer to like elements throughout the description of the figures.

(20) It should be understood that when an element or layer is referred to as being “on,” “connected to,” “coupled to,” or “covering” another element or layer, it may be directly on, connected to, coupled to, or covering the other element or layer or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly connected to,” or “directly coupled to” another element or layer, there are no intervening elements or layers present. Like numbers refer to like elements throughout the specification. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(21) It should be understood that, although the terms first, second, third, etc. may be used herein to describe various elements, regions, layers and/or sections, these elements, regions, layers, and/or sections should not be limited by these terms. These terms are only used to distinguish one element, region, layer, or section from another region, layer, or section. Thus, a first element, region, layer, or section discussed below could be termed a second element, region, layer, or section without departing from the teachings of example embodiment.

(22) Spatially relative terms (e.g., “beneath,” “below,” “lower,” “above,” “upper,” and the like) may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It should be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the term “below” may encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

(23) The terminology used herein is for the purpose of describing various example embodiments only and is not intended to be limiting of example embodiments. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, and/or elements, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements and/or groups thereof.

(24) Example embodiments are described herein with reference to cross-sectional illustrations that are schematic illustrations of example embodiment. As such, variations from the shapes of the illustrations are to be expected. Thus, example embodiment should not be construed as limited to the shapes of regions illustrated herein but are to include deviations and variations in shapes.

(25) When the words “about” and “substantially” are used in this specification in connection with a numerical value, it is intended that the associated numerical value include a tolerance of $\pm 10\%$ around the stated numerical value, unless otherwise explicitly defined. Moreover, when the terms “generally” or “substantially” are used in connection with geometric shapes, it is intended that precision of the geometric shape is not required but that latitude for the shape is within the scope of the disclosure. Furthermore, regardless of whether numerical values or shapes are modified as “about,” “generally,” or “substantially,” it will be understood that these values and shapes should be construed as including a manufacturing or operational tolerance (e.g., $\pm 10\%$) around the stated numerical values or shapes.

(26) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which example embodiment belong. It will be further understood that terms, including those defined in commonly

used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(27) FIG. 1A is a perspective view of an electronic unit according to at least one example embodiment. FIG. 1B is a detailed, perspective view of a filtering system of the electronic unit of FIG. 1A according to at least one example embodiment.

(28) Referring to FIGS. 1A and 1B, in at least one example embodiment, an electrical unit, such as an electronic unit **100**, includes a filtering system **105**. For example, the electronic unit **100** may include a radio frequency device, such as a remote radio head or remote radio unit. The filtering system **105** may be configured to be removably coupled or attached to the electronic unit **100**. For example, at least one side of the electronic unit **100** may include at least one opening configured to receive the filtering system **105**, as will be discussed in more detail below with respect to FIGS.

8A-8B. Thus, according to at least one example embodiment, the filtering system **105** may be positioned on at least one side or side surface of the electronic unit **100**. In one example, the electronic unit **100** may be a remote radio head (RRH) or other radio access network (RAN) apparatus or device. Example embodiments should not, however, be limited to this example.

(29) In at least one example embodiment, the filtering system **105** is configured to prevent and/or at least partially or substantially prevent liquid and/or solid particles from entering the electronic unit **100** while allowing air to enter and exit the electronic unit **100**.

(30) FIG. 2 is a perspective view of the filtering system of FIG. 1B according to at least one example embodiment.

(31) Referring to FIG. 2, in at least one example embodiment, the filtering system **105** includes a base **200** and a cover **205**. The cover **205** is configured to be coupled to the base **200**. The base **200** and the cover **205** may be referred to as a filtering apparatus **210**. The filtering system **105** may also include an air vent **215**. For example, the filtering apparatus **210** is configured to receive at least a portion of the air vent **215**.

(32) FIG. 3 is an exploded view of the filtering system of FIG. 2 according to at least one example embodiment.

(33) Referring to FIG. 3, in at least one example embodiment, the base **200** may define one or more first holes **220** and the cover **205** may define one or more second holes **225**. The one or more first holes **220** may be positioned adjacent a periphery of the base **200**. The one or more second holes **225** may be positioned adjacent a periphery of the cover **205**. The one or more first holes **220** may be configured to align with the one or more second holes **225**. In at least one example embodiment, one or more attachment devices **230** may be configured to be inserted through the one or more first holes **220** and the one or more second holes **225** to secure the cover **205** to the base **200**. For example, the one or more attachment devices **230** may include one or more screws, bolts, pins, other attachment or fixing means, or the like.

(34) In at least one example embodiment, the cover **205** includes one or more recesses **235** corresponding with the one or more second holes **225**. The one or more recesses **235** allow a top surface of each of the one or more attachment devices **230** to be flush with a top of the cover **205** when inserted through the one or more second holes **225** and the one or more first holes **220**.

(35) In at least one example embodiment, the base **200** defines a cavity **238** and an opening **240**. The cavity **238** and the opening **240** of the base **200** may be configured to receive at least a portion of the air vent **215**. For example, the air vent **215** may include a first portion **245** and a second portion **250** at an end of the first portion **245**. In at least one example embodiment, the first portion **245** of the air vent **215** may be configured to extend through the opening **240** of the base **200** and the cavity **238** of the base **200** may be configured to receive the second portion **250** of the air vent **215**, as will be discussed in greater detail with respect to FIG. 7B.

(36) In at least one example embodiment, the filtering system **105** includes a gasket **255**. The gasket **255** may be positioned around the first portion **245** of the air vent **215** adjacent the opening

240 of the base **200**, as shown in FIGS. 7B and 8A-8B. For example, the gasket **255** may be configured to be positioned between the base **200** and the electronic unit **100** when the filtering system **105** is inserted into or attached to the electronic unit **100**. In at least one example embodiment, the gasket **255** is configured to create a fluid seal between the filtering apparatus **210** and the electronic unit **100**.

(37) FIG. 4A is a top, perspective view of the base of the filtering system of FIGS. 2-3 according to at least one example embodiment. FIG. 4B is a bottom, perspective of the base of FIG. 4A according to at least one example embodiment. FIG. 4C is a side, cross-section view of the base of FIG. 4A according to at least one example embodiment.

(38) Referring to FIGS. 4A-4C, in at least one example embodiment, the base **200** includes a bottom wall **400**, a first sidewall **405** extending from a perimeter of the bottom wall **400**, a support wall **410**, and a second sidewall **415**. The first sidewall **405** and the second sidewall **415** extend from respective ends of the support wall **410**. For example, the first sidewall **405** and the second sidewall **415** may be perpendicular or substantially perpendicular to the support wall **410**. A length of the first sidewall **405** may be greater than a length of the second sidewall **415**. In other example embodiments, the length of the second sidewall **415** may be equal to or greater than the length of the first sidewall **405**.

(39) In at least one example embodiment, the first sidewall **405**, the support wall **410**, and the second sidewall **415** define a groove **420**. The groove **420** may extend at least partially about a perimeter of the first sidewall **405**. For example, the first holes **220** of the base **200** may extend through the support wall **410** and the groove **420** may extend between each of the first holes **220**.

(40) In at least one example embodiment, the bottom wall **400** of the base **200** defines the opening **240** and the bottom wall **400** and the first sidewall **405** of the base **200** define the cavity **238**. The opening **240** is in fluid communication with the cavity **238**. The base **200** is configured to receive the air vent **215** such that the first portion **245** extends through the opening **240** and the second portion **250** is within the cavity **238**. In at least one example embodiment, the bottom wall **400** of the base **200** defines a bottom recess **430**. The bottom recess **430** may surround the opening **240** and have a shape corresponding to a shape of the second portion **250** of the air vent **215**. For example, the second portion **250** of the air vent **215** may comprise a polygonal shape having at least three sides. In one example, the second portion **250** may have six sides as shown in FIG. 3. The bottom recess **430** of the base **200** may include the same number of sides as the second portion **250** such that at least a portion of the second portion **250** is configured to be seated within the bottom recess **430**. In other example embodiments, the second portion **250** of the air vent **215** and the bottom recess **430** may have a circular shape.

(41) In at least one example embodiment, the first sidewall **405** may define a plurality of side recesses **435**. The plurality of side recesses **435** may be positioned about the interior surface of the first sidewall **405** within the cavity **238**. The plurality of side recesses **435** may create a plurality of corresponding extensions, such as side extensions **440**, extending from an exterior surface of the first sidewall **405**, as shown in at least FIG. 4B. The plurality of side recesses **435** may comprise one or more different sizes, widths, and/or shapes. In other example embodiments, each of the plurality of side recesses **435** may be the same size and shape. Each of the plurality of side recesses **435** may create a gap between the second portion **250** of the air vent **215** and the interior surface of the first sidewall **405** when the air vent **215** is positioned within the base **200** such that fluid (e.g., air) may flow between the first sidewall **405** and the second portion **250**, as will be discussed in greater detail with respect to FIGS. 8A-8B. As shown in FIG. 4A, for example, the plurality of side recesses **435** may be spaced apart evenly from one another about the perimeter of the interior surface of the first sidewall **405** within the cavity **238**. However, example embodiments should not be limited to this example. Rather, the plurality of side recesses may be spaced apart unevenly in other embodiments. Similarly, the side extensions **440**, extending from an exterior surface of the first sidewall **405**, may be spaced apart evenly or unevenly from one another.

(42) FIG. 5A is a top, perspective view of the cover of the filtering system of FIGS. 2-3 according to at least one example embodiment. FIG. 5B is a bottom, perspective view of the cover of FIG. 5A according to at least one example embodiment.

(43) Referring to FIGS. 5A and 5B, in at least one example embodiment, the cover **205** includes a top wall **500** having a first side **505** and a second side **510**. The cover **205** also includes a third sidewall **515** extending from a perimeter of the second side **510** of the top wall **500**. The third sidewall **515** may be perpendicular or substantially perpendicular to the top wall **500**. As mentioned above with respect to FIG. 3, the top wall **500** may define one or more second holes **225** configured to receive the one or more attachment devices **230** and may define one or more recesses **235** surrounding or partially surrounding the one or more second holes **225**.

(44) In at least one example embodiment, the cover **205** includes at least one spacer **520** extending from the second side **510** of the top wall **500**. The at least one spacer **520** may be configured to define one or more cavities or compartments between the base **200** and the cover **205** when the cover **205** is coupled to the base **200**, as will be discussed with respect to at least FIG. 7A. The at least one spacer **520** may include a protrusion **525** and/or a plurality of ribs **530** extending from the second side **510** of the top wall **500**. For example, the protrusion **525** may extend from a center or central portion of the second side **510** of the top wall **500**. In at least one example embodiment, the first side **505** of the top wall **500** defines a recess corresponding to the protrusion **525**, such as a top recess **535**. The plurality of ribs **530** may extend from the second side **510** of the top wall **500** between the protrusion **525** and the third sidewall **515**. As shown in the example illustrated in FIGS. 5A and 5B, the protrusion **525** and the top recess **535** may have a hexagonal shape, and each of six ribs **530** may extend from a vertex of the hexagonal shape towards the third sidewall **515**. Example embodiments should not, however, be limited to this example.

(45) FIG. 6A is a side, perspective view of the air vent of the filtering system of FIGS. 2-3 according to at least one example embodiment. FIG. 6B is a side, cross-section view of the air vent of FIG. 6A according to at least one example embodiment.

(46) Referring to FIGS. 6A and 6B, in at least one example embodiment, the first portion **245** of the air vent **215** may have a cylindrical or substantially cylindrical shape. The second portion **250** of the air vent **215** may have a polygonal shape. In one example, the second portion **250** may have a polygonal shape with a plurality of sides of differing lengths. In another example, the second portion may have a cylindrical or substantially cylindrical shape. A circumference or diameter of the second portion **250** may be greater than a circumference or diameter of the first portion **245**. In other example embodiments, the circumference or diameter of the first portion **245** may be the same or substantially the same as the circumference or diameter of the second portion **250**.

(47) In at least one example embodiment, the second portion **250** of the air vent **215** includes a top part **600** and a bottom part **605**. The bottom part **605** is at an end of the first portion **245** of the air vent **215**. The top part **600** may be separated from the bottom part **605** by a plurality of supports **610** to define a space **615** between the top part **600** and the bottom part **605**. For example, the plurality of supports **610** may be spaced about an outer perimeter of the second portion **250** between the top part **600** and the bottom part **605**. The second portion **250** may also define a plurality of side openings **620** between each of the plurality of supports **610** in fluid communication with the space **615**.

(48) In at least one example embodiment, the first portion **245** of the air vent **215** defines a channel **625**. The channel **625** may comprise a cylindrical or substantially cylindrical shape and extend along a length of the first portion **245**. The channel **625** is in fluid communication with the space **615** between the top part **600** and the bottom part **605**. The channel is also configured to be in fluid communication with the electronic unit **100**. In at least one example embodiment, the air vent **215** includes a membrane filter configured to prevent or substantially prevent the passage of liquid and/or solid particles into the electronic unit **100**. For example, the membrane filter is configured to prevent or substantially prevent the passage of liquid and/or solid particles into the channel **625** and

the electronic unit **100** from the space **615** while allowing air to flow between the space **615** and the channel **625**. In at least one example embodiment, at least a portion of the bottom part **605** of the air vent includes the membrane filter. In other example embodiments, the membrane filter is fluidly coupled to or within the channel **625**.

(49) In at least one example embodiment, an exterior surface of the first portion **245** of the air vent **215** comprises a plurality of threads **630**. The plurality of threads **630** may be configured to secure the air vent **215**, and thereby the filtering system **105** including the base **200** and the cover **205**, to the electronic unit **100**, as will be described with respect to FIGS. **8A-8B**.

(50) FIG. **7A** is a side, cross-section view of the base and the cover of FIG. **2** according to at least one example embodiment. FIG. **7B** is a side, cross-section view of the filtering system of FIG. **2** according to at least one example embodiment.

(51) Referring to FIGS. **7A** and **7B**, the base **200** and the cover **205** form the filtering apparatus **210** of the filtering system **105**. The cover **205** is configured to be coupled to the base **200**. For example, the cover **205** defines a portion of the cavity **238** when coupled to the base **200**. The cover **205** is also configured to cover the air vent **215**, as shown in FIG. **7B**. For example, the air vent **215** may be inserted into the opening **240** and seated within the cavity **238** of the base **200** prior to attaching the cover **205**. The cover **205** may then be coupled to the base **200**. The cover **205** may be secured to the base **200** by the one or more attachment devices **230** such that the base **200**, the cover **205**, and the air vent **215** are secured (e.g., removably secured) together, as shown in FIG. **7B**.

(52) With reference to FIG. **7A**, the third sidewall **515** of the cover **205** is configured to be parallel or substantially parallel with the second sidewall **415** of the base **200** when the cover **205** is coupled to the base **200**. A circumference or diameter of the cover **205** is greater than a circumference or diameter of the base **200** such that a gap **700** is defined between the second sidewall **415** of the base **200** and the third sidewall **515** of the cover **205**. The gap **700** is configured to be in fluid communication with the cavity **238**. In at least one example embodiment, the gap **700**, the second sidewall **415**, and the third sidewall **515** are cylindrical or substantially cylindrical. However, example embodiments should not be limited to this example.

(53) In at least one example embodiment, at least a portion of the at least one spacer **520** is configured to contact a top surface of the support wall **410** of the base **200**. For example, at least a portion of each of the ribs **530** are configured to contact a top surface of the support wall **410** of the base **200**. The protrusion **525** may be configured to contact a top surface of the air vent **215**, such as a top surface of the top part **600**, as shown in FIG. **7B**. In at least one example embodiment, the second side **510** of the top wall **500** of the cover **205**, each of the plurality of ribs **530**, the protrusion **525**, and the support wall **410** of the base **200** define a plurality of compartments **705**. Each of the plurality of compartment **705** are fluidly coupled between the gap **700** and the cavity **238** of the base **200**. For example, each of the plurality of compartments **705** may be fluidly coupled to the cavity **238** via the plurality of side recesses **435**.

(54) Still referring to FIG. **7A**, the ambient environment may be in fluid communication with the interior of the filtering apparatus **210** through the gap **700**, the one or more compartments **705**, and the cavity **238**.

(55) Referring to FIG. **7B**, when the air vent **215** is within the filtering apparatus **210**, the ambient environment may be in fluid communication with the air vent **215** via the gap **700**, the one or more compartments **705**, the plurality of side recesses **435**, the side openings **620** and the space **615** of the second portion **250**, and the channel **625**.

(56) FIG. **8A** is a detailed, cross-section view of the filtering system and the electronic unit of FIGS. **1A-1B** showing a first fluid pathway according to at least one example embodiment. FIG. **8B** is a detailed, cross-section view of the filtering system and the electronic unit of FIGS. **1A-1B** showing a second fluid pathway according to at least one example embodiment.

(57) Referring to FIGS. **8A** and **8B**, in at least one example embodiment, the filtering system **105** is

configured to be removably coupled to the electronic unit **100**. For example, the filtering system **105** may be coupled to the electronic unit **100** by threaded engagement using the threads **630** of the air vent **215**. The threads **630** may be configured to engage an interior surface of a cavity **810** defined by the electronic unit **100**. In other example embodiments, the filtering system **105** may be removably coupled to the cavity **810** of the electronic unit **100** by a friction fit, snap fit, or other engagement mechanism or means.

(58) As mentioned above with respect to FIG. **3**, the filtering system **105** may include a gasket **255**. In at least one example embodiment, the gasket **255** surrounds the first portion **245** of the air vent **215** adjacent the second portion **250**. For example, the gasket **255** contacts at least a portion of a lower surface of the bottom part **605** of the second portion **250** of the air vent **215**. As shown in FIGS. **8A-8B**, the gasket **255** is between the filtering system **105** and the electronic unit **100** and configured to create a fluid seal.

(59) In at least one example embodiment, the filtering system **105** is configured to create at least two fluid pathways. With reference to FIG. **8A**, an interior surface of the third sidewall **515** of the cover **205**, the second side **510** of the top wall **500** of the cover **205**, an exterior surface of the second sidewall **415** of the base **200**, and a top surface of the support wall **410** define a first fluid pathway **800**.

(60) In at least one example embodiment, the first fluid pathway **800** is configured to direct liquid and/or solid particles that enter the filtering system **105** from the ambient environment away from the electronic unit **100**. For example, liquid and/or solid particles may enter the filtering system **105** through the gap **700**. The first fluid pathway **800** directs the liquid and/or solid particles from the gap **700** into the compartment **705** and then towards the opposite end or bottom of the filtering system **105** by gravity such that the liquid and/or solid particles exit the filtering system **105** via one of the compartments **705** and the gap **700** at the opposite end of the filtering system **105**. The liquid and/or solid particles within the first fluid pathway **800** may be directed toward the bottom of the filtering system **105** through each of the compartments **705** defined by the plurality of ribs **530** and the protrusion **525**.

(61) In at least one example embodiment, the first fluid pathway **800** prevents or substantially prevents liquid and/or solid particles, such as dust, from entering the air vent **215** so the air vent does not get blocked. For example, if liquid and/or solid particles accumulate at or within the air vent **215**, the air vent **215** may become blocked and air flow in and out of the electronic unit **100** may be inhibited. Such blockage may create pressure in the electronic unit **100**, such as a vacuum, which can cause the electronic unit **100** to malfunction.

(62) With reference to FIG. **8B**, an interior surface of the third sidewall **515** of the cover **205**, the second side **510** of the top wall **500** of the cover **205**, an exterior surface of the second sidewall **415**, a top surface of the support wall **410**, and an interior surface of the first sidewall **405** define at least a portion of a second fluid pathway **805**. The second fluid pathway **805** may be an air pathway in fluid communication with the electronic unit **100**. For example, the second fluid pathway **805** may be configured to direct air from the ambient environment through the air vent **215** and into the electronic unit **100**. The second fluid pathway **805** defines a labyrinth that air can enter and traverse, but liquid and/or solid particles cannot. For example liquid and/or solid particles may drop within the filtering system **105**, such as through the first fluid pathway **800**, before reaching the air vent **215**.

(63) In at least one example embodiment, the second fluid pathway **805** directs air through the gap **700**, the compartment **705**, and the side recesses **435** towards the air vent **215**. The air may enter the air vent **215** through the side openings **620**, travel through the space **615** and the channel **625**, and enter the electronic unit **100** through the cavity **810**. As discussed above, the air vent **215** may include a membrane filter configured to prevent liquid and/or solid particles from entering the channel **625** and/or the electronic unit **100**. For example, at least a portion of the bottom part **605** of the air vent **215** may comprise the membrane filter. In other example embodiments, the membrane

filter may be within the channel **625** and/or the cavity **810**. In still other example embodiments, the membrane filter may be within the filtering apparatus **210**, such as within the filtering apparatus **210** shown in FIG. 7B. For example, the membrane filter may be positioned within the cavity **238** and/or coupled to an interior surface of the base **200** or the cover **205**. In such example embodiments, the membrane filter may take the place of the air vent **215**.

(64) In at least one example embodiment, the second fluid pathway **805** may also flow in a reverse direction. For example, air may exit the electronic unit **100** through the air vent **215** and the filtering apparatus **210** of the filtering system **105**. This allows the electronic unit **100** to breathe such that air may enter and exit the electronic unit **100**.

(65) FIG. 9 is a detailed, cross-section view of another filtering system and electronic unit according to at least one example embodiment.

(66) Referring to FIG. 9, in at least one example embodiment, a filtering system **903** may be integrated into an electronic unit **900**. For example, the electronic unit **900** may include at least one protrusion **905** adjacent a cavity **910** defined by a side of the electronic unit **900**. The cavity **910** is configured to be in fluid communication with an interior of the electronic unit **900**. The at least one protrusion **905** may be a cylindrical or substantially cylindrical protrusion surrounding the cavity **910**. In other example embodiments, the at least one protrusion **905** may comprise a first protrusion adjacent the cavity **910** and a second protrusion adjacent the cavity **910** opposite the first protrusion.

(67) In at least one example embodiment, the cavity **910** is configured to receive at least a portion of a filter, such as the air vent **215** discussed above with respect to FIGS. 3 and 6A-8B. In at least one example embodiment, the at least one protrusion **905** defines a recessed space **915** adjacent the cavity **910**. A circumference, width or diameter of the recessed space **915** may be greater than a circumference, width or diameter of the cavity **910**. The cavity **910** may be configured to receive the first portion **245** of the air vent **215** and the recessed space **915** may be configured to receive the second portion **250** of the air vent **215**. For example, the second portion **250** of the air vent **215** may be seated within the recessed space **915** such that a bottom surface of the bottom part **605** contacts a surface of the electronic unit **900** surrounding the cavity **910** and a top surface of the top part **600** of the air vent **215** is flush with an exterior surface of the at least one protrusion **905**.

(68) In at least one example embodiment, a cover **920** is coupled to the at least one protrusion **905**. The cover includes a top wall **925** and a sidewall **930** extending from a perimeter of the top wall **925**. The sidewall **930** may be configured to couple the cover **920** to the at least one protrusion **905**. In at least one example embodiment, the cover **920** includes at least one engagement member **935** extending from an end of the sidewall **930** opposite the top wall **925**. For example, the at least one engagement member **935** may extend from the end of the sidewall **930** perpendicular to the sidewall **930** and parallel to the top wall **925**. The at least one engagement member **935** may be configured to engage an exterior portion of the at least one protrusion **905**, such as in a snap-fit or other engagement mechanism, to secure the cover **920** to the electronic unit **900**. The at least one engagement member **935** may be a single, cylindrical member extending about a perimeter of the sidewall **930**. In other example embodiments, the at least one engagement member **935** may be a plurality of discrete members extending from a perimeter of the sidewall **930**.

(69) In at least one example embodiment, at least one protrusion **940** extends from an interior surface of the top wall **925** of the cover **920**. The at least one protrusion **940** is configured to extend towards the air vent **215** and the electronic unit **900** when the cover **920** is coupled to the electronic unit **900**. The cover also defines at least one recess **945** adjacent the at least one protrusion **940**.

(70) In at least one example embodiment, the filtering system **903** defines at least two fluid pathways. For example, the filtering system **903** defines a first fluid pathway configured to direct liquid and/or solid particles that enter the filtering system **903** away from the air vent **215** and out of the filtering system **903**, such as the first fluid pathway **800** described above with respect to FIG. 8A. The first fluid pathway may be defined by an interior surface of the sidewall **930**, an exterior

surface of the at least one protrusion **905**, and the interior surface of the top wall **925** of the cover **920**. The liquid and/or solid particles may be directed downward along the interior surface of the top wall **925** by gravity, along the at least one protrusion **940**, and through the at least one recess **945**. Because the liquid and/or solid particles are directed along the interior surface of the cover **920**, the liquid and/or solid particles may be prevented from entering the electronic unit **900** through the air vent **215**. The liquid and/or solid particles may exit through a bottom portion of the cover **920**.

(71) In at least one example embodiment, the filtering system **903** defines a second fluid pathway configured to allow air to enter and exit the electronic unit **900**, similar to the second fluid pathway **805** described above with respect to FIG. **8B**. The second fluid pathway may be defined by an interior surface of the sidewall **930**, an exterior surface of the at least one protrusion **905**, at least a portion of the interior surface of the top wall **925** of the cover **920**, and the recessed space **915**. Air may enter and exit the air vent **215** and the electronic unit **900** via the recessed space **915** in a similar or analogous manner as described above with respect to FIG. **8B**. For example, from the recessed space **915**, air may enter the air vent **215** through the side openings **620** such that the air is directed through the space **615**, through a membrane filter, through the channel **625**, through the cavity **910**, and into the electronic unit **900**.

(72) Example embodiments have been disclosed herein, however, it should be understood that other variations may be possible. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure, and all such modifications as would be obvious to one skilled in the art are intended to be included within the scope of the following claims.

Claims

1. An apparatus for attaching to an electronic unit, comprising: a base including, a bottom wall defining an opening, a first sidewall extending from a perimeter of the bottom wall, a support wall, and a second sidewall, the first sidewall and the second sidewall extending from respective ends of the support wall such that a groove is defined between the first sidewall, the support wall, and the second sidewall; and a cover configured to be coupled to the base, the cover including, a top wall including a first side and a second side, a third sidewall extending from a perimeter of the second side of the top wall, and at least one spacer extending from the second side of the top wall and configured to define a cavity between the base and the cover; wherein an interior surface of the third sidewall of the cover, the second side of the top wall of the cover, an exterior surface of the second sidewall, and a top surface of the support wall define a first fluid pathway; and an interior surface of the third sidewall of the cover, the second side of the top wall of the cover, an exterior surface of the second sidewall, a top surface of the support wall, and an interior surface of the first sidewall define a second fluid pathway, and wherein the first fluid pathway is a liquid pathway configured to direct liquid away from the electronic unit; and the second fluid pathway is an air pathway in fluid communication with the electronic unit.
2. The apparatus of claim 1, wherein the at least one spacer includes, a protrusion extending from a center of the second side of the top wall; and a plurality of ribs extending from the second side of the top wall between the protrusion and the third sidewall, the plurality of ribs configured to contact at least a top surface of the support wall of the base.
3. The apparatus of claim 1, further comprising an air vent within the opening of the bottom wall.
4. The apparatus of claim 3, wherein the air vent is configured to prevent passage of liquid and solid particles into the electronic unit.
5. The apparatus of claim 3, wherein the cover is configured to cover the air vent and the at least one spacer is configured to contact at least a portion of a top surface of the air vent.
6. The apparatus of claim 1, wherein the top wall of the cover defines at least one hole, the at least one hole configured to receive at least one attachment device for securing the cover to the base.

7. The apparatus of claim 6, wherein the at least one attachment device comprises one or more screws.
 8. The apparatus of claim 1, wherein the first sidewall and the second sidewall are perpendicular to the support wall.
 9. The apparatus of claim 1, wherein a length of the first sidewall is greater than a length of the second sidewall.
 10. A filtering system for an electronic unit, the filtering system comprising: a base including, a bottom wall defining an opening, a first sidewall extending from a perimeter of the bottom wall, a support wall, and a second sidewall, the first sidewall and the second sidewall extending from respective ends of the support wall such that a groove is defined between the first sidewall, the support wall, and the second sidewall; an air vent including, a first portion configured to extend through the opening in the bottom wall of the base, the first portion defining a channel, and a second portion at an end of the first portion and in fluid communication with the channel; and a cover configured to be coupled to the base and cover the air vent, the cover including, a top wall including a first side and a second side, a third sidewall extending from a perimeter of the second side of the top wall, and at least one spacer extending from the second side of the top wall and configured to define a cavity between the base and the second side of the cover; wherein an interior surface of the third sidewall of the cover, the second side of the top wall of the cover, an exterior surface of the second sidewall, and a top surface of the support wall define a first fluid pathway; and an interior surface of the third sidewall of the cover, the second side of the top wall of the cover, an exterior surface of the second sidewall, a top surface of the support wall, an interior surface of the first sidewall, the second portion of the air vent, and the channel of the first portion of the air vent define a second fluid pathway, and wherein the first fluid pathway is a liquid pathway configured to direct liquid away from the electronic unit; and the second fluid pathway is an air pathway in fluid communication with the electronic unit.
 11. The filtering system of claim 10, wherein the at least one spacer includes, a protrusion extending from a center of the second side of the top wall and configured to contact a top surface of the air vent, and a plurality of ribs extending from the second side of the top wall between the protrusion and the third sidewall, the plurality of ribs configured to contact at least a top surface of the support wall of the base.
 12. The filtering system of claim 10, wherein: the second portion of the air vent includes a top part and a bottom part; the bottom part at the end of the first portion of the air vent; and the top part separated from the bottom part by a plurality of supports extending between the top part and the bottom part.
 13. The filtering system of claim 12, wherein a space defined between the top part and the bottom part is in fluid communication with the channel.
 14. The filtering system of claim 10, further comprising: a gasket surrounding at least a portion of the first portion of the air vent; and wherein an exterior surface of the first portion of the air vent includes a plurality of threads.
 15. The filtering system of claim 10, wherein the air vent is configured to prevent passage of liquid and solid particles into the electronic unit.
 16. The filtering system of claim 10, wherein the top wall of the cover defines at least one hole, the at least one hole configured to receive at least one attachment device for securing the cover to the base.
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